





ABSTRACT

CricketAI: An Intelligent Virtual Cricket Coach Using Pose Estimation and Retrieval-Augmented Generation

Personalised technique feedback is difficult for most amateur cricketers to obtain: qualified coaches are scarce, recorded video needs expert interpretation, and general-purpose chatbots are not grounded in curated coaching knowledge. This paper presents CricketAI, a web-based virtual cricket coach that integrates single-camera pose analysis, a retrieval-augmented generation (RAG) assistant and structured skill lessons in one application. The video-analysis pipeline decodes an uploaded batting video with OpenCV, estimates 33 body landmarks per frame with MediaPipe Pose, derives geometric measurements such as joint angles, and converts them through rule-based evaluation into a technical score, a confidence value and a per-metric report. The coaching assistant embeds user questions with BAAI bge-base-en-v1.5, retrieves candidate passages from a ChromaDB store of approximately 1,002 cricket knowledge chunks, reorders them with a BGE cross-encoder reranker and generates grounded answers with a locally hosted Qwen3:4B model, so user queries are not sent to external AI services. The system is implemented with a React–TypeScript frontend and a FastAPI–MongoDB backend secured with JSON Web Tokens. Functional validation confirmed end-to-end execution of authentication, dashboard, video upload and retrieval, and coaching-query operations, and a complete analysis of a 1,654-frame clip recorded at 30.03 frames per second. We separate geometric measurements from coaching judgements, discuss the limits of monocular pose estimation, and define an evaluation protocol covering pose-measurement error, retrieval Recall, expert-rated answer quality, latency and usability as the basis for validation.

Keywords: cricket coaching, human pose estimation, MediaPipe, retrieval-augmented generation, reranking, large language models, sports analytics

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